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|----------|-----------------------------------------------------------------------|
| Ex.no:7b | <b>DATA VISUALIZATION USING TABLEAU-<br/>VEHICLE ANALYSIS DATASET</b> |
|          |                                                                       |

### Aim

To connect, analyze, visualize, and create interactive dashboards using Tableau with the **Vehicle Analysis Dataset** for effective analysis of vehicle prices, usage, fuel types, transmission types, seller types, and ownership.

### Software Requirements

- Tableau Desktop / Tableau Public
- Microsoft Excel
- CSV Dataset
- Windows Operating System

### Theory

Tableau is a powerful Business Intelligence (BI) and Data Visualization tool used to create interactive dashboards and reports.

The **Vehicle Analysis Dataset (VAD)** contains information about used vehicles, including:

- Car Name
- Manufacturing Year
- Selling Price
- Present Price
- Kilometers Driven
- Fuel Type
- Seller Type
- Transmission
- Number of Previous Owners

Tableau allows users to analyze vehicle prices, usage patterns, fuel preferences, transmission types, and ownership characteristics through interactive visualizations.

### Tableau Architecture

#### Vehicle Analysis Dataset

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#### Data Connection

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#### Worksheets

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## Visualizations

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## Dashboard

↓

## Story

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### Learning Outcomes

After completing this experiment, students will be able to:

- ✓ Connect the Vehicle Analysis Dataset to Tableau
- ✓ Create worksheets using dimensions and measures
- ✓ Generate different charts and visualizations
- ✓ Analyze vehicle prices
- ✓ Analyze vehicle usage based on kilometers driven
- ✓ Apply filters and parameters
- ✓ Create calculated fields
- ✓ Build interactive dashboards
- ✓ Create stories for vehicle data analysis

### Sample Dataset

#### Vehicle Analysis Dataset

| Car_Name | Year | Selling_Price | Present_Price | Kms_Driven | Fuel_Type | Transmission |
|----------|------|---------------|---------------|------------|-----------|--------------|
| ritz     | 2014 | 3.35          | 5.59          | 27000      | Petrol    | Manual       |
| sx4      | 2013 | 4.75          | 9.54          | 43000      | Diesel    | Manual       |
| ciaz     | 2017 | 7.25          | 9.85          | 6900       | Petrol    | Manual       |
| wagon r  | 2011 | 2.85          | 4.15          | 5200       | Petrol    | Manual       |
| swift    | 2014 | 4.60          | 6.87          | 42450      | Diesel    | Manual       |

*The actual VAD contains a much larger number of records and columns.*

## **Algorithm 1 – Connecting Data Source**

### **Step 1**

Open **Tableau Desktop**.

### **Step 2**

Select **Connect**.

### **Step 3**

Choose **Text File**.

### **Step 4**

Browse and select the Vehicle Analysis Dataset.

### **Step 5**

Open the dataset.

### **Step 6**

Verify the imported fields such as:

- Year
- Car\_Name
- Selling\_Price
- Present\_Price
- Kms\_Driven
- Fuel\_Type
- Seller\_Type
- Transmission
- Owner

## **Procedure**

### **Tableau**



### **Connect**



### **Excel / Text File**



### **Vehicle Analysis Dataset**

↓

**Open**

↓

**Verify Data**

**Output**

**Vehicle Analysis Dataset loaded into Tableau successfully.**

**Algorithm 2 – Creating Worksheet**

**Step 1**

Open Sheet 1.

**Step 2**

Drag **Fuel\_Type** to Columns.

**Step 3**

Drag **Selling\_Price** to Rows.

**Step 4**

Change Selling\_Price aggregation to **Average**.

**Step 5**

Select an appropriate visualization.

**Output**

**Worksheet created successfully.**

**Interpretation**

The worksheet displays the average selling price of vehicles according to their fuel type.

**Algorithm 3 – Bar Chart Creation**

**Objective**

To compare the average selling price of vehicle based on fuel type.

**Step 1**

Drag **Fuel\_Type** → Columns.

## Step 2

Drag **Selling\_Price** → Rows.

## Step 3

Select **Bar Chart** from Show Me.

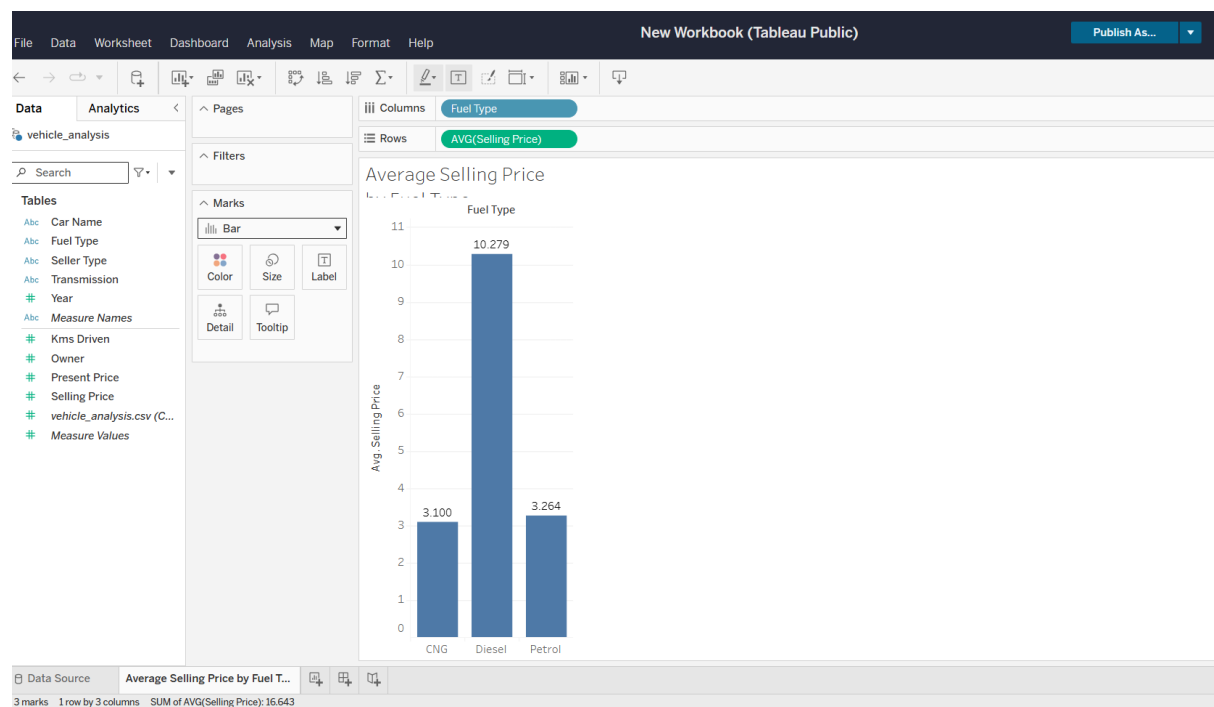
## Step 4

Change Selling\_Price to **Average**.

## Step 5

Enable **Show Mark Labels**.

## Output Visualization



## Interpretation

The bar chart helps compare the average selling prices of vehicles using different fuel types.

## Algorithm 4 – Pie Chart Creation

### Objective

To analyze the distribution of vehicles based on fuel type.

## Step 1

Drag **Fuel\_Type** to the view.

## Step 2

Select **Pie Chart** from Show Me.

## Step 3

Drag **Car\_Name** to **Angle**

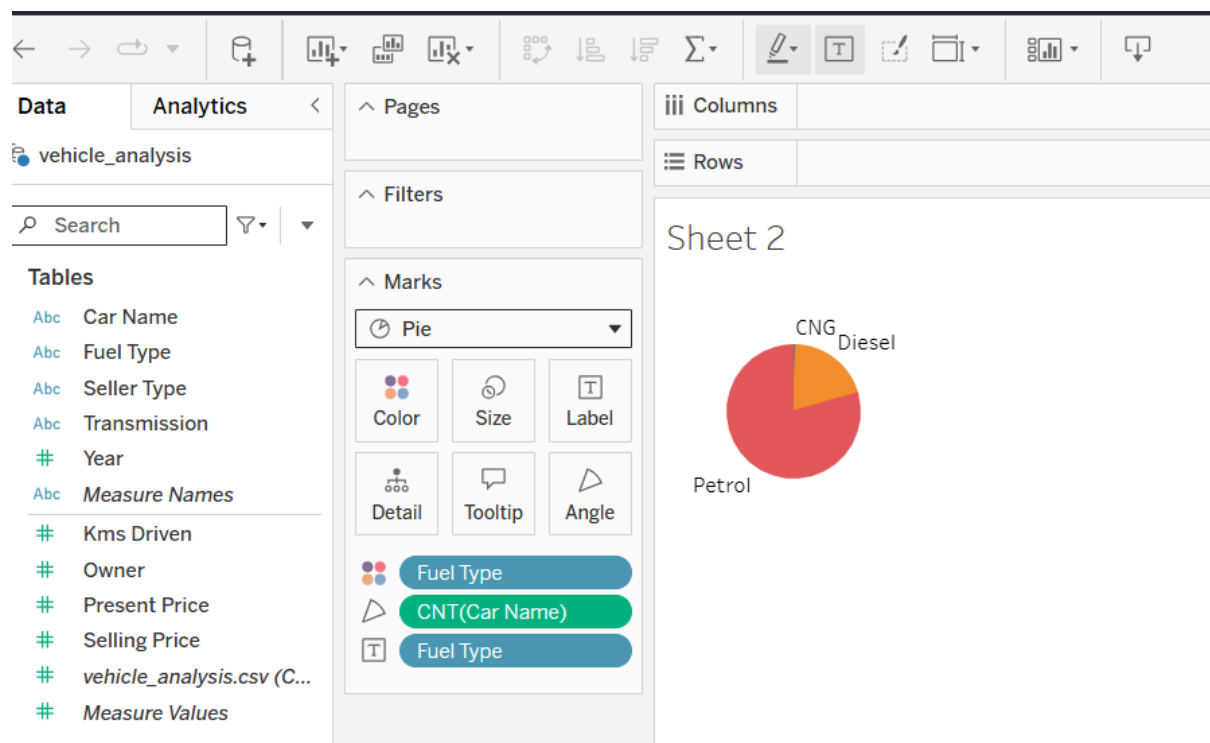
## Step 4

Drag **Fuel\_Type** to **Color**.

## Step 5

Enable **Show Labels**.

## Output Visualization



## Interpretation

The pie chart shows the proportion of vehicles belonging to different fuel types such as Petrol, Diesel, and CNG.

## Algorithm 5 – Line Chart Creation

### Objective

To analyze the trend vehicle selling prices over the years.

## Step 1

Drag **Year** → Columns.

## Step 2

Drag **Selling\_Price** → Rows.

## Step 3

Change Selling\_Price aggregation to **Average**.

## Step 4

Select **Line Chart**.

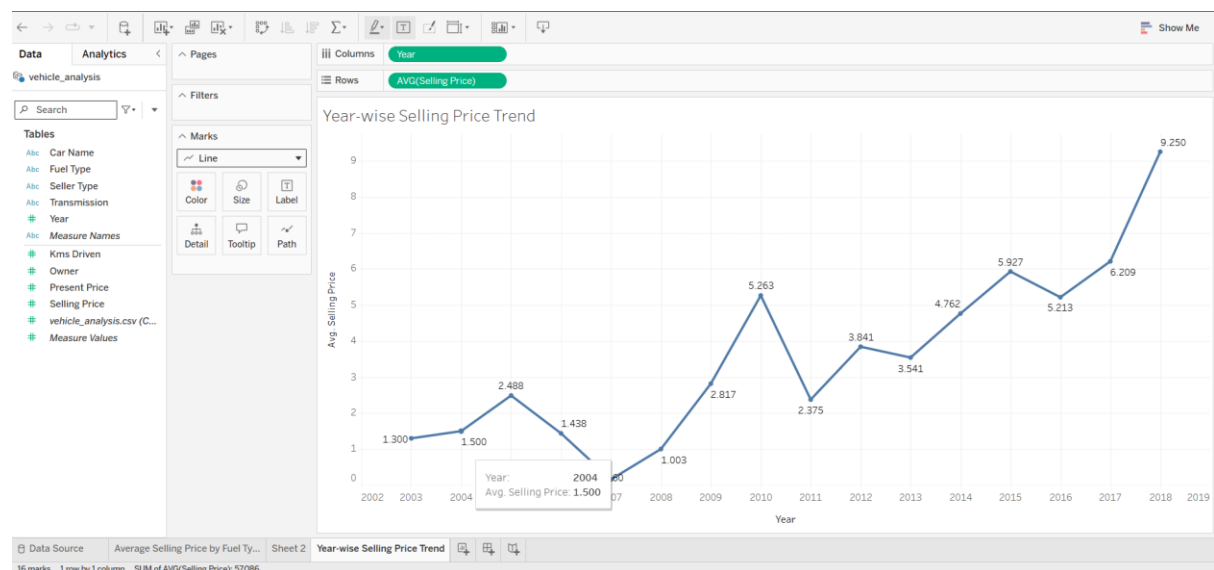
## Step 5

Apply appropriate formatting.

## Step 6

Analyze the trend.

## Output Visualization



## Interpretation

The line chart shows how the average selling price of vehicles changes across different manufacturing years.

It helps identify:

- Increasing price trends
- Decreasing price trends
- High/Low-price years
- Changes in vehicle resale value

## Algorithm 6 – Creating Calculated Field

### Objective

To calculate the **Vehicle Age** from the manufacturing year.

### Step 1

Select:

**Analysis → Create Calculated Field**

### Step 2

Enter the field name:

**Vehicle Age**

### Step 3

Enter the formula:

[2026] + [Year]

### Step 4

Click **OK**.

### Output

| Car_Name | Year | Vehicle Age |
|----------|------|-------------|
| ritz     | 2014 | 12          |
| sx4      | 2013 | 13          |
| ciaz     | 2017 | 9           |
| wagon r  | 2011 | 15          |

### Interpretation

The calculated field determines the approximate age of each vehicle using its manufacturing year.

## Algorithm 7 – Applying Filters

### Objective

To analyze vehicles based on a specific fuel type.

### Step 1

Drag **Fuel\_Type** to Filters.

### Step 2



Select the required fuel type.

### **Step 3**

Apply the filter.

### **Step 4**

Observe the updated charts.

### **Example**

**Filter: Fuel\_Type = Diesel**

### **Output**

The visualization displays only vehicles using diesel fuel.

### **Interpretation**

Filters allow users to focus on a particular category of vehicles and perform detailed analysis.

## **Algorithm 8 – Creating Parameter**

### **Objective**

To display vehicles based on a user-defined minimum selling price.

### **Step 1**

Select:

### **Create Parameter**

### **Step 2**

Specify the parameter name:

**Minimum Fatalities**

### **Step 3**

Set the data type as:

**Float**

### **Step 4**

Set the required minimum value.

### **Step 5**

Create a calculated field using the parameter.

### **Formula**

```
[Selling_Price] >= [Minimum Selling Price]
```

### **Example**

**Minimum Selling Price = 5**

### **Output**

The visualization displays vehicles having a selling price of **5 or more**.

### **Interpretation**

Parameters allow users to dynamically control the vehicle analysis based on a selected selling-price value.

## **Algorithm 9 – Dashboard Creation**

### **Objective**

To create an interactive **Vehicle Analysis Dashboard**.

### **Step 1**

Create multiple worksheets.

### **Step 2**

Select **Dashboard** → **New Dashboard**.

### **Step 3**

Drag the required worksheets into the dashboard.

### **Step 4**

Arrange the charts appropriately.

### **Step 5**

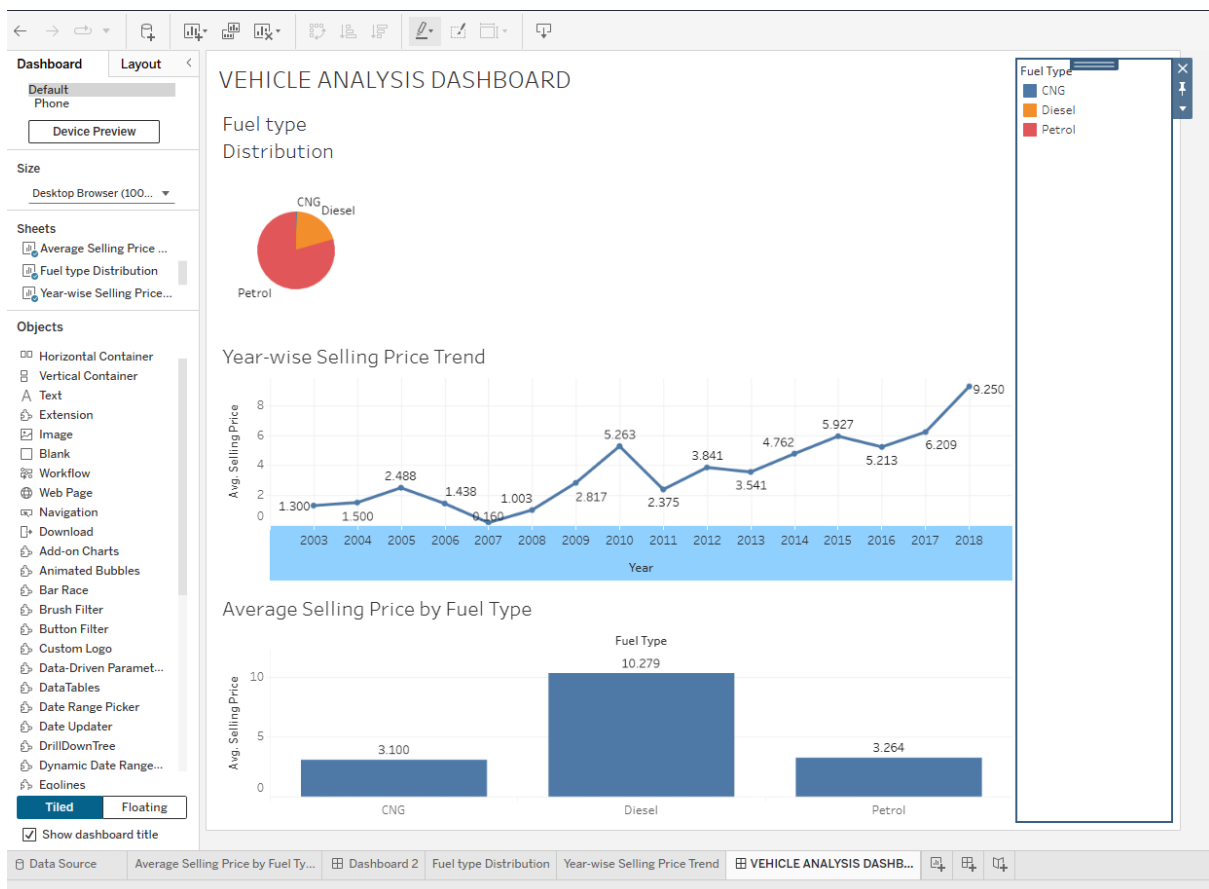
Add filters for:

- Year
- Fuel\_Type
- Seller\_Type
- Transmission
- Owner

### **Step 6**

Add appropriate titles.

## Dashboard Analysis



## Total Vehicles

Number of vehicle records in the dataset

## Average Selling Price

Average selling price of vehicles

## Average Present Price

Average current/present price of vehicles

## Average Kilometer Driven

Average distance driven by vehicles

## Fuel Type Distribution

Distribution of vehicles according to fuel type

## Transmission Distribution

Distribution of manual and automatic vehicles

### **Interpretation**

The dashboard provides a consolidated view of vehicle characteristics, prices, usage, fuel types, and transmission types and allows users to interactively analyze the data using filters.

### **Algorithm 10 – Story Creation**

#### **Objective**

To present the analysis of vehicle analysis through a sequence of visual insights.

#### **Step 1**

Create the required worksheets.

#### **Step 2**

Create the dashboard.

#### **Step 3**

Select **Story** → **New Story**.

#### **Step 4**

Add the dashboard and required worksheets as story points.

#### **Step 5**

Arrange the story points in a logical sequence.

#### **Step 6**

Present the findings.

### **Example Story**

#### **Story Point 1**

##### **Overall Vehicle Analysis**

Displays total vehicles, average selling price, present price, and kilometers driven.

#### **Story Point 2**

##### **Price Analysis**

Shows the relationship and variation between vehicle selling price and present price.

#### **Story Point 3**

## **Fuel Type Analysis**

Shows the distribution of vehicles based on Petrol, Diesel, and CNG.

## **Story Point 4**

## **Vehicle Usage Analysis**

Analyzes vehicles based on kilometers driven.

## **Story Point 5**

## **Transmission Analysis**

Compares vehicles with manual and automatic transmission.

## **Story Point 6**

## **Final Conclusions**

Summarizes the major patterns and observations obtained from the Vehicle Analysis Dataset.

## **Applications**

### **Automobile Industry**

Vehicle price and market analysis.

### **Used Car Market**

Analysis of resale prices and vehicle demand.

### **Sales Analysis**

Comparison of selling prices across different vehicle categories.

### **Customer Analysis**

Understanding fuel and transmission preferences.

### **Business Intelligence**

Interactive vehicle sales dashboards and reports.

### **Data Analytics**

Exploratory analysis of vehicle characteristics and pricing trends.

## **Result**